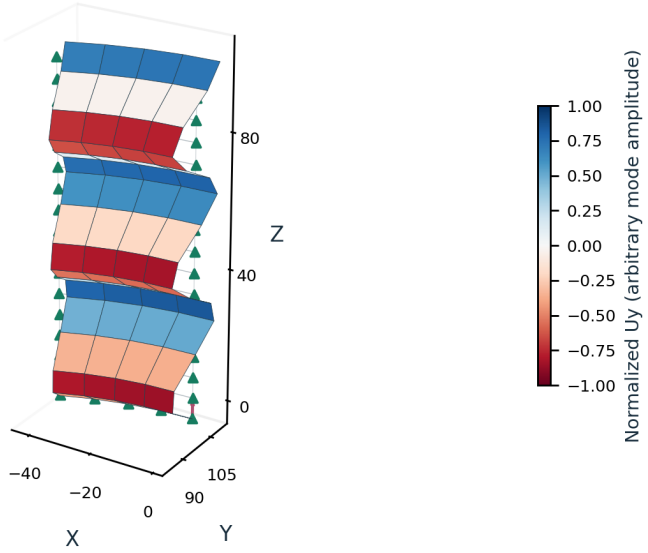
Cylindrical strip: axial buckling

BA_cylinder_axial_buckling | SOL 105 | Analytical screening; finer extraction remains partial

Thin cylinder R=100 m, full length 200 m, t=1 m, E=210 GPa, nu=0.3; one 22.5-degree circumferential sector over half-length.

Geometry / mesh	18 nodes; 8 elements. Thin cylinder R=100 m, full length 200 m, t=1 m, E=210 GPa, nu=0.3; one 22.5-degree circumferential sector over half-length.
Material / section	E=2.1e+11, nu=0.3; shell thickness=1. SI: m, N, kg, s, Pa
Boundary conditions	Cylindrical-frame symmetry and axial midpoint constraints; 1 Pa axial compression. The strip imposes axisymmetric deformation.
Applied load	Uniform 1 Pa reference axial compression; consistent edge forces sum to 39.270 in +Z on the strip end.

Mode 1 | load factor = 1.2778e+09



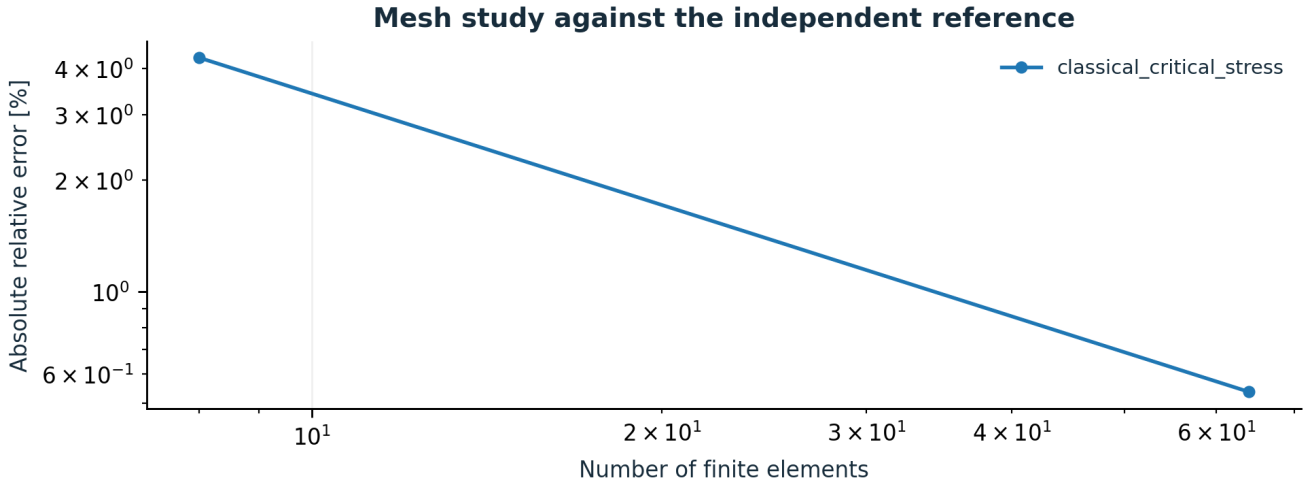
Shape amplified x12; grey = original. Eigenvector sign and amplitude are arbitrary.

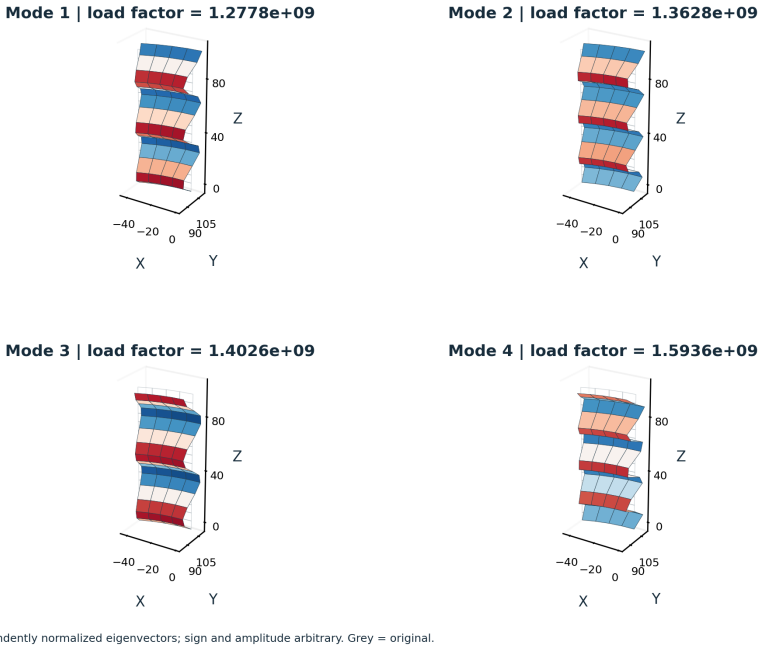
Recorded JFEM result; deformation is amplified, mode amplitudes/signs arbitrary. Green triangles mark constrained nodes; magenta arrows mark available point loads. Finest companion shown.

Recorded numerical checks

Quantity [unit]	JFEM	Reference	Error	Assessment
Critical stress (lambda x 1 Pa) [GPa]	1.3254	1.271	4.2809%	PASS
preload uz node 1 [m]	4.7716e-10	4.7716e-10	1.8085e-06%	REGRESSION
N16x04 refined response [GPa]	1.2778	1.271	0.53713%	PASS
32x8 first retained root (unqualified) [GPa]	1.2765	1.271	0.43454%	PARTIAL

Evidence, convergence and interpretation





Independently normalized eigenvectors; sign and amplitude arbitrary. Grey = original.

First available modes. Red/blue indicate opposite signed components; each mode is normalized independently.

The qualified strip refinement is 16x4 (first-root error 0.537%). Finer 32x8 extraction attempts failed completeness checks: the retained five-of-six result is published as PARTIAL, with no certified global critical-root claim.

Finite strip boundary conditions differ from an unlimited ideal cylinder. The remaining discrepancy must not automatically be attributed entirely to mesh error.

The supplied input omitted SPC selection from the buckling subcase. This rerun explicitly adds SPC=1 to that subcase, preserving the intended constraints. The original failed attempt is retained in the audit record.

Scope: Only the first qualified buckling root is compared with ideal theory. The 32x8 spectrum remains unqualified; this is a finite axisymmetric screening model, not an experimental capacity prediction.

Reference and reproduction

Brush and Almroth, Buckling of Bars, Plates and Shells (1975), classical perfect-cylinder theory.

Hilburger, M.W. (2020). Buckling of Thin-Walled Circular Cylinders. NASA/SP-8007-2020/REV 2. Ideal theory is distinct from imperfection-sensitive test capacity. https://ntrs.nasa.gov/citations/20205011530

Open the recorded .jfem file with Boxer: Project > Open JFEM results. For older eigenvalue files choose modal or buckling results when requested. The input deck and the recorded run receipt accompany the figures.

Run: 2026-10-19 | Solver revision: a70f997 | Full precision values: results.json / metrics.csv